Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “velocity” and “speed” to explain your model or result. Sentence starters are optional.

velocity: rate of position change, with direction

speed: how fast something moves, without direction

slope: vertical change divided by horizontal change on a graph

01 Bring back an earlier idea

Start at 3 m, move to 10 m, then finish at 5 m.
Draw the route.
Find distance, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt.
Δ means change; Δt is elapsed time. Position-time slope gives velocity.

Worked reasoning: A cart moves from 1 m to 7 m in 3 s at steady velocity. The position-time slope is (7-1)/3=2 m/s.

Your turn: Sketch the graph through the two points.
Draw and label the rise and run used for its slope.

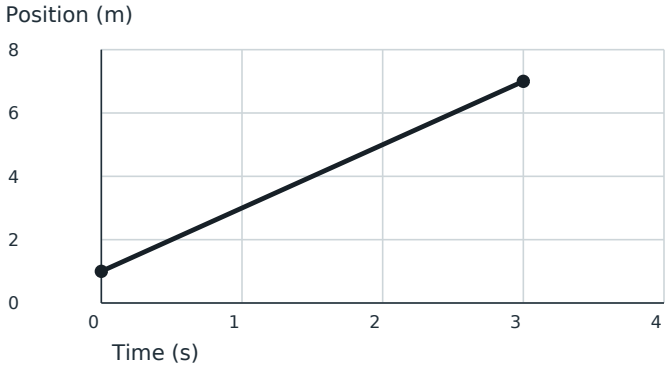
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Read the motion picture

The graph shows a cart moving right. Read its position at 0 s and 3 s.
Draw an arrow showing its direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

04 Find the rate

Use the graph in 3. Position change = ____ - ____ = ____ m. Time = ____ s. Velocity = ____ m/s to the ____.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Velocity and position graphs

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 A pause

A position-time graph stays at 5 m from 4 s to 6 s. Is the cart moving?
Explain what stays the same.

Support: The cart is ____ because its position ____.

Sentence starter: My claim is … . The diagram/data show …, so … .

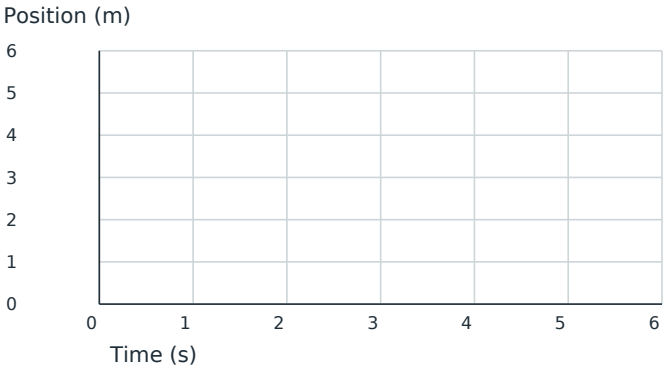
Marzano check - circle: 1 2 3

06 Draw a walk with a pause

Plot the supplied points and connect neighbors with straight lines.
Label moving right, stopped, and moving left.
Use the blank graph.

Support: Data: t = 0, 2, 4, 6 s; x = 1, 5, 5, 1 m.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

07 Distance or displacement?

For the walk in 6, find distance traveled and displacement. Why is average velocity zero even though the person walked?

Sentence starter: My claim is … . The diagram/data show …, so … .

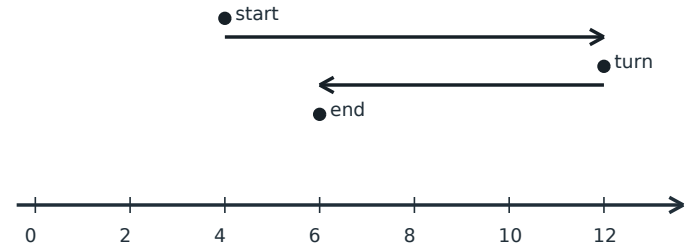
Marzano check - circle: 1 2 3

08 Choose a method

Start at 4 m, move to 12 m, then finish at 6 m.
Draw the route.
Find distance, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

A cart goes from 9 m to 3 m in 5 s at steady velocity.
Find the velocity with direction.
Sketch the shape of its position-time graph.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

10 STRETCH: Apply it another way

Draw two straight position-time graphs with different starting positions but the same velocity of 2 m/s.
Explain the feature they share.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3